

DogChat: A Pet-centered Smart Collar Prototype based on Large Language Models and Wechat

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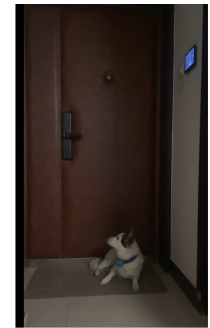
A. Smart Collar



B. Eating Food



C. Weather Observation



D. Doorbell Scenario

Figure 1: Different scenarios

ABSTRACT

This study explores how large language models (LLMs) based smart collars can enable pets to express themselves and interact with their owners through social media. We proposed *DogChat*: a pet-centered prototype system that captures real-time visual and auditory data in first person from pets through smart collars and utilizes LLMs for pets' behavior and emotion analysis. The system operates in three phases: Pet Profile Construction, Daily Experience Reconstruction, and Behavior Learning Integration. Through the WeChat platform, pets can actively share their status and alert to abnormal events, or respond passively to owner inquiries.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile devices.

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KEYWORDS

Wearable Devices; Animal-Centered Design; Large Language Models

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1 INTRODUCTION

In current pet care practices, traditional pet monitoring technologies help pet owners acquire basic behaviors and physiological states of their pets. However, these technologies are often limited to simple data collection from a third-person perspective, such as activity monitoring and basic health tracking, while the first-person perspective of pets' behavior and emotional states is usually ignored[14, 15]. This makes it challenging for pet owners to deeply understand their pets' behavior and emotions when they are not around[9]. In addition, vision-based technologies have also been used to monitor and identify behavioral changes in pets such as dogs and cats [11]. Another study introduced a wearable device using a single Inertial Measurement Unit (IMU) that can monitor

and analyze dog behaviors to improve their health and address various issues related to pet care [1].

With the development of Large Language Models (LLMs) and advanced computer vision technologies, we now have the opportunity to overcome these limitations[3, 16]. These technologies can process and interpret complex behavioral data, not only enhancing the accuracy of data analysis but also enabling the transformation of pets' behaviors and emotional states into detailed descriptions that pet owners can understand. Particularly, the introduction of LLMs opens up new possibilities for in-depth interpretation and interaction with non-verbal animal behaviors, signaling a qualitative leap in the field of pet care.

In this work, we proposed *DogChat*: a smart collar prototype system integrated with visual and auditory sensors, embedded with the latest LLM technology, to achieve real-time monitoring and in-depth analysis of behaviors and emotions in a first-person perspective of the pets. Through this system, we converted pets' behaviors and emotional states into text and images and pushed them to their owners' social media platforms. We chose WeChat to implement the prototype, where pets can engage in both active expressions and passive queries, aiming to enhance emotional connections and interactive experiences between pets and their owners.

2 RELATED WORKS

2.1 Technologies for Detecting Pet Behavior and Emotion

Existing research extensively applies computer vision and sensor technologies for monitoring and detecting pet behavior. Studies using the YOLOv3-Tiny model and Raspberry Pi achieved real-time monitoring of cat behaviors with an average accuracy of 98% [2, 6]. Another study developed an LSTM-based method using accelerometer and gyroscope data to detect and classify 10 activities of dogs with good accuracy and high performance, showing promise for real-time monitoring[7]. Additionally, pet emotion recognition technologies are also evolving. For instance, research indicates that dogs' emotional reactivity is influenced by the quality of the dog-owner relationship, with factors such as emotional closeness, shared activities, and perceived costs playing a significant role in modulating dogs' arousal levels and behaviors. Another study found that owners believe dogs can feel more emotions than cats and use different behavioral signs to identify various emotions in dogs and cats[12]. These technologies provide foundational support for the smart collar system in our research.

2.2 Pets in Social Media

In recent years, the expression of pets on social media has attracted widespread attention. Existing research primarily focuses on how pet owners manage and use their pets' accounts on social media. Studies show that dog owners tend to seek advice and share experiences on pet-oriented social networks, while cat owners use these platforms more to build communities[5]. Additionally, research has analyzed the interaction between pet-level and human-level networks in online pet social networks, mainly focusing on network structures and relationships between users rather than autonomous pet expression[4]. The motivations and strategies for managing pet

social media accounts have also been extensively explored, highlighting the importance of these accounts in online marketing[8]. Although users of pet Instagram accounts present their "fur babies" to bring joy to their followers, these expressions are mainly controlled by pet owners and lack mechanisms for autonomous pet expression[10].

Large Language Models (LLMs) and role-playing techniques have significantly advanced intelligent interaction and user experience research. LLMs simulate human conversation and provide personalized responses, enhancing user engagement and satisfaction in education, healthcare, and entertainment. Role-playing, widely used in training and education, promotes situational learning and skill development[17]. Combining LLMs with role-playing creates realistic interactive scenarios, enhancing immersion and learning outcomes[13]. This integration offers new methods for constructing intelligent interactive systems, with the potential for breakthrough advancements in various applications.

Our research aims to overcome these limitations by integrating a smart collar system with visual and auditory sensors and leveraging the latest LLM technology to achieve real-time monitoring and autonomous expression of pets' behaviors and emotional states, providing a richer and more interactive experience.

3 DESIGN AND DEVELOPMENT

In response to the limitations of traditional pet care methods, we developed an intelligent pet monitoring system that integrates advanced sensors and artificial intelligence. The overall architecture of the system implementation is shown in Fig2. The design goals of the system include:

Real-time Data Capture and Transmission: Utilizing the low-power ESP32-CAM module integrated into a smart collar, the system continuously monitors pet behavior and transmits data in real time.

Analyzing Pet Behavior with Large Language Models: The GLM-4V model on the server side analyzes the collected data, providing detailed insights into pet behavior patterns and emotional states.

Social Media Interaction: The system implements automatic content posting and real-time interaction features through WeChat, enhancing the emotional connection between pets and their owners.

3.1 Real-time Data Capture and Transmission

In our system, the integration of the ESP32-CAM module within the smart collar is crucial for real-time capture of visual and auditory data from pets. This module is equipped with an OV2640 camera, which has a resolution of 2 megapixels and supports a maximum image quality of 1600x1200. To conserve battery power, in this study we are using a method where we capture one photo every 2 seconds. It can record video at up to 20 fps, ensuring every detail of pet movements is accurately captured. Additionally, to comprehensively capture the sounds of pets, a highly sensitive I2S MEMS microphone is embedded, capable of clearly picking up sounds within a range of approximately 10 meters, suitable for various environments. The module is shown as Fig3.

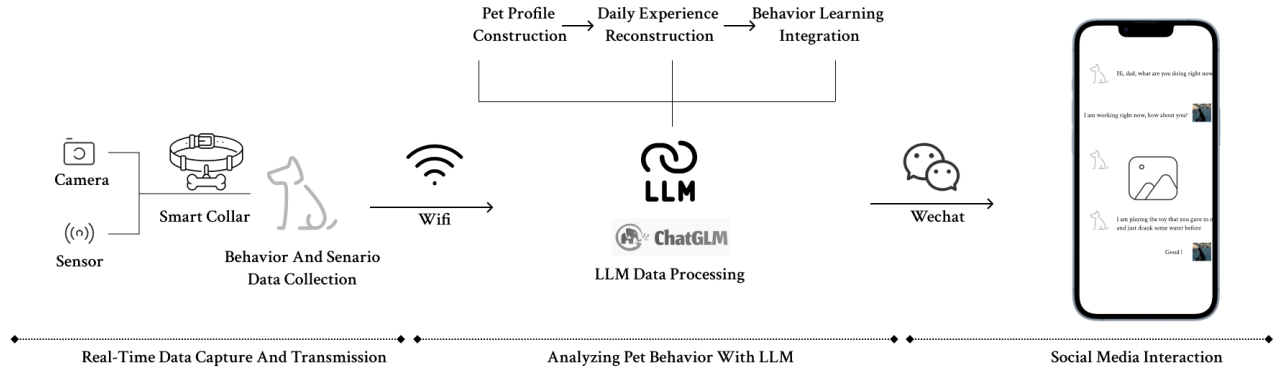


Figure 2: Overall Framework

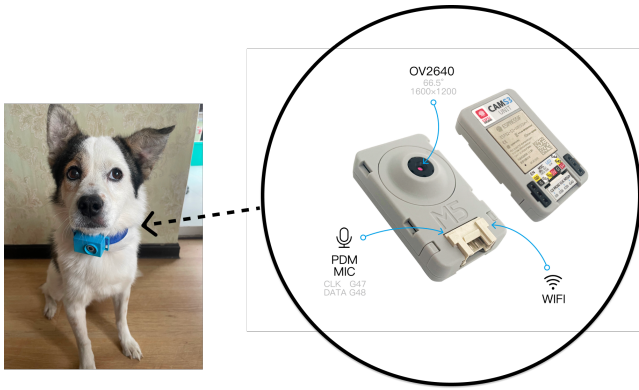


Figure 3: Data Collection Module

The entire smart collar is powered by a 3.7V lithium battery, ensuring the device's portability and long-duration usability. Data is transmitted in real-time to the server via Wi-Fi technology supporting the 802.11b/g/n standard and is protected using WPA/WPA2 encryption protocols, ensuring the security and integrity of the transmission. This precise data capture and transmission technology provides pet owners with real-time, detailed monitoring information, enhancing the system's practicality and security.

3.2 Analyzing Pet Behavior with Large Language Models

To enable our intelligent agent to effectively learn from real-world data, we have designed a three-phase approach: *Pet Profile Construction*, *Daily Experience Reconstruction*, and *Behavior Learning Integration*. This phased approach ensures precise adaptation and optimization from basic data collection to complex behavior analysis and real-time prediction. The agent design framework is shown as Fig4.

Pet Profile Construction: In this phase, the main task is to collect basic information and historical behavior data of the pet, including: 1) basic information: breed, age, gender, health records, etc; and 2) behavior Logs: detailed records of daily activities, dietary habits, sleep patterns, emotional responses, etc. This data provides

the initial background information for the model, helping it to understand the fundamental characteristics of the pet.

Daily Experience Reconstruction: In this phase, we analyze and process the collected data from the Profile Collection phase, converting it into "experiences" that the model can learn from. Specific steps include: 1) data extraction: extracting key behavior and emotional data from the collected profiles; and 2) scenario reconstruction: integrating this data into specific behavior and emotional scenarios, such as typical behaviors and emotional states during play, eating, and resting. These experience data form the foundation for the model to understand the pet's daily behaviors and emotional responses.

Behavior Learning Integration: In this phase, the reconstructed experience data is uploaded to the trained model, allowing it to learn and optimize based on these detailed experiences. Specific steps include: 1) role formation: using the rich experience data, the model can develop an understanding of the pet's character and feelings, making its behavior more personalized; 2) personalized understanding: the model learns from detailed behavioral experiences to more accurately understand the pet's personality and behavior patterns; and 3) behavior prediction: through continuous experience learning, the model can predict the pet's future behaviors and emotional changes, helping pet owners better understand and care for their pets.

3.3 Social Media Interaction

WeChat is one of the most popular social media platforms in China, offering a wide range of communication and social networking features, including instant messaging, Moments, official accounts, and mini-programs. To facilitate interaction between pets and their owners, our system leverages these WeChat functionalities through technical integration, automating pet social activities. Chathistoy is as shown Fig5.

Technical Implementation: Our system's implementation relies on Python programming language and the itchat library, an open-source interface for personal WeChat accounts, which allows for programmatically managing WeChat client interactions.

Automatic Login and Message Management: Using the itchat library, the system can automatically log into WeChat, send, and receive messages. It simulates user operations, managing pet owners'

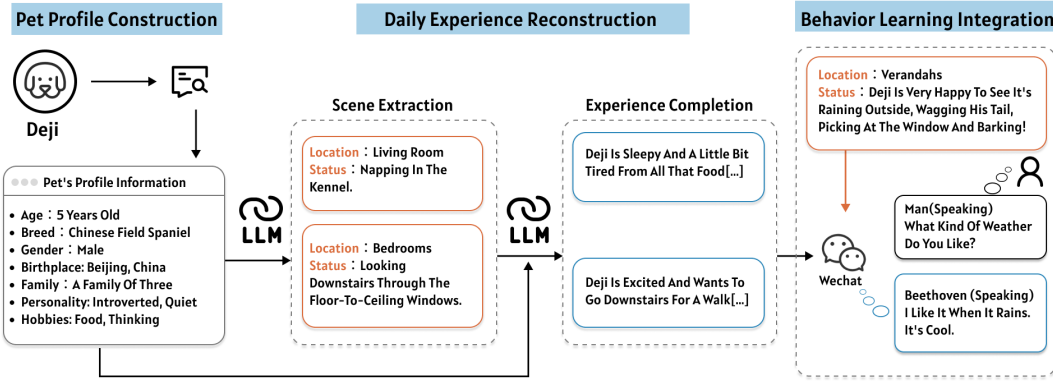


Figure 4: LLM agent design framework

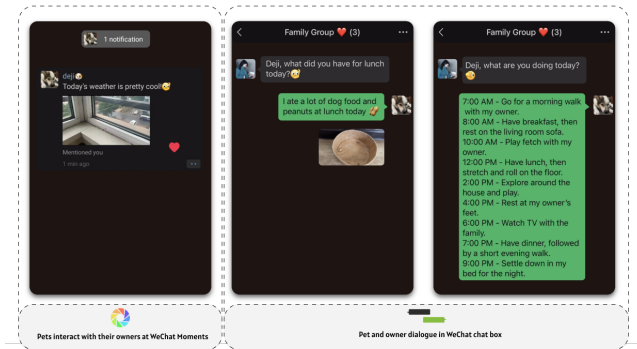


Figure 5: Scenario chat

inquiries and replying to messages in real-time. Content Posting: The system can automatically generate text and images suitable for social media and use the itchat library to post this content directly to the pet owner's Moments or send it to specific contacts. Data Encryption and Security: All data sent and received through WeChat is encrypted, ensuring communication security and the privacy of user data.

Active Expression: The system enables pets to actively communicate through automatically updated posts based on analyzed data of pet behavior and emotions: Dynamic Sharing: The system analyzes the daily activities and emotional changes of pets and automatically converts this information into graphical updates, posting on the pet owner's WeChat Moments. Emergency Messaging: In the event of an emergency, such as health problems or abnormal behavior, the system immediately notifies the pet owner, allowing for timely intervention.

Passive Expression: The passive expression feature allows pet owners to actively inquire about their pet's status: Daily Activity Inquiry: Pet owners can send messages to inquire about their pets' daily activities, such as walking, playing, or resting times. Health Status Inquiry: The system provides detailed reports on the pet's

health and dietary conditions, allowing pet owners to stay informed about their pets' well-being at any time.

4 DISCUSSION AND FUTURE WORKS

This research has developed a smart collar that is connected to a large language model and social media platforms. Through image and voice information, it provides comprehensive and dynamic monitoring of pet behavior and emotions, supporting social interaction between owners and their pets. This innovative design offers new possibilities for pet behavior analysis and emotional interaction. However, the system involves the transmission of sensitive pet video and audio data, which requires attention to user privacy and data security concerns. Therefore, it is necessary to widely collect user feedback and continuously optimize the interface interactions and functional design to ensure the system truly meets user needs.

In the future, we will optimize and expand this pet monitoring system from multiple perspectives. First, we will conduct in-depth user experience research to understand the impact of the new interaction on pet owners and pets, and then optimize the system functions to meet user needs. At the same time, we will continue to improve the system's ability to interpret complex pet emotional expressions, combined with computer vision and other cutting-edge technologies, to enhance user experience with more accurate emotional analysis.

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